



Triple Tres (333)

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AI / LLM

MCP ProtocolAgentic Loops

LLM Orchestration

MCP PrimitivesMulti-Agent

Claude APITool Use API

RAG · pgvectorSSE Transport

n8n FlowsLangfuseMastra

HITL PatternsMake · Zapier

BACKEND / INFRA

TypeScriptNode.jsPython

PostgreSQLSQLiteDocker

RailwayREST APIsGitLinux

FRONTEND

Next.jsReactTailwind

TypeScriptshadcn/ui

EDUCATION & CERTS

- Model Context Protocol: Advanced Topics
Anthropic · Oct 2025
- Claude Code in Action
Anthropic · Sep 2025
- Introduction to Model Context Protocol
Anthropic · Sep 2025
- AI Fluency: Framework & Foundations
Anthropic · 2025
- n8n Self-Hosted for Enterprises
n8n · Nov 2025
- Building Scalable Agentic Systems
DataCamp · 2025
- B.A. Graphic Design
UAA · 2013

LANGUAGES

SpanishNative

EnglishProfessional — technical documentation, proposals, and async collaboration with US-based teams

AI SOFTWARE ENGINEER · MCP SERVER DESIGN · LLM AGENTIC SYSTEMS

AI engineer building **production-grade agent systems** in TypeScript. Designed and shipped a **43-tool MCP server**, a multi-tier Claude orchestration layer, and the full fiscal-compliance platform on top of it — from zero, with strict Zod validation, **6,000+ tests**, and end-to-end observability. I build infrastructure that holds up in production, document in **English-first environments** by default, and want to bring that into an engineering team.

EXPERIENCE

- AI Engineer · DISAI** (*independent*) Jun 2025 – Present
- 43-tool MCP server.** Architected and deployed a production MCP server (TypeScript/Node.js, Railway) across all three primitives — **Tools** for CFDI 4.0 fiscal ops, **Resources** for live SAT-catalog injection (batch-loaded per agent domain), **SSE + HTTP Streamable** dual transport, multi-tenant CSD management, EFOS/EDOS blacklist sync, PAC stamping. **Benchmarked at ~1.2ms median tool dispatch with a 0% error rate under sustained synthetic load.** MCPTypeScriptRailway
 - 3-tier agentic orchestration.** Haiku router (classifies 10 fiscal domains + query intent in ~100ms) → 10 Sonnet DomainAgents (Anthropic native tool_use, agentic loops up to 6 iterations, parallel tool calls, self-correction on failure) → **Expert Registry** with scoped tool allowlists (5–8 per domain) and live MCP Resource injection. Claude APITool UseMulti-Agent
 - Hallucination control by design.** Pre-load SAT catalog Resources into each agent's context before the first LLM call — eliminates a whole class of hallucinated catalog codes without RAG query overhead. MCP Resourcespgvector
 - Observability & cost control.** Instrumented the orchestrator with Langfuse end-to-end — every LLM call, tool invocation, model selection, token cost, and streaming event traced, with per-session cost attribution and failure triage without log digging. LangfuseTracing
 - Human-in-the-loop surfaces.** Built a compliance dashboard and EFOS monitor as Next.js reviewer interfaces with one-click approval flows and MCP-UI in-chat apps, wired into the agent pipeline as the human approval layer for irreversible operations. Next.jsHITLMCP-UI
 - Streaming React frontend.** Next.js 16 + shadcn/ui chat with an SSE event pipeline (activity log, thinking steps, skeleton states for 30–60s async waits), compliance dashboard, CSD status, onboarding, per-call savings tracking. Next.jsReact 19Tailwind
 - Engineering discipline.** 6,000+ tests, 91%+ coverage, CI/CD, conventional commits, schema validation at every boundary — solo-built to a standard that's safe to hand to a team. CI/CDZodVitest

- Core Contributor / Lead Developer · RetailDAO** Jun 2024 – Jan 2026
- Real-time analytics terminal.** Built and shipped a "Bloomberg-style" terminal for the community's research operations — aggregated Federal Reserve, on-chain, and exchange data over dual WebSockets, with rate-limiting, caching, and graceful degradation for low-latency uptime. ReactNode.jsRedisDocker
 - Distributed team & audits.** Collaborated with a distributed contributor team; led internal smart-contract audits and built a token-distribution visualization platform that reduced investor risk.

- Crypto Research Analyst · Coinsider** Mar 2024 – Sep 2025
- Data-driven intelligence.** Delivered 25+ peer-reviewed analytics reports (1,200+ pages) — SWOT analysis, investor-risk models, and macro forecasting for Web3 ecosystems, translating dense technical data into clear writing that reached **70k+ views**.

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KEY PROJECTS

MCP SDK · TypeScript
Claude API · Tool Use
Next.js 16 · React 19
Langfuse · HITL
n8n · Docker
PostgreSQL · pgvector
Railway · Coolify

KEY PROJECTS

SAT-MCP Server

43-tool MCP server for Mexican fiscal compliance. Full primitive set: tools, resources, prompts, elicitations, tasks + 8 MCP-UI mini-apps for in-chat fiscal rendering. SSE + HTTP Streamable. CFDI 4.0 lifecycle, EFOS/EDOS sync, PAC stamping, multi-tenant CSD. 5,500+ tests · 91%+ coverage.

TypeScript MCP SDK Node.js
Railway SQLite

DISAI_Conta – 3-Tier Agent System

Haiku router + 10 domain DomainAgents (Sonnet, native tool_use, 6-iter agentic loops) + Expert Registry with scoped allowlists and live MCP Resource injection. Langfuse full-stack tracing. HITL compliance dashboard. Streaming SSE UI.

Next.js 16 Claude API Tool Use
Langfuse shadcn/ui

n8n AI Automation Stack

Self-hosted n8n with direct MCP client integration for SAT tool execution. Specialized sub-agents per CFDI type, context-bundle architecture, 400+ available integrations. Replaced Zapier/Make for an ~80% recurring-cost reduction.

n8n MCP LLM Agents Docker

Homelab AI Infrastructure

PostgreSQL 16 + pgvector for semantic SAT catalog search. Self-hosted Gitea, static IPs, Samba share. Powers the MCP Resource injection pipeline in production.

PostgreSQL pgvector Linux Gitea
Docker

Mastra + Langfuse Stack Evaluation

Evaluated Mastra framework for HITL patterns, agent workflows, and MCP connections. Identified tradeoffs vs. custom harness: Soniox TTS incompatibility, shallower Langfuse A/B testing depth. Production stack uses a custom orchestrator with pure Langfuse cloud for full observability and cost-attribution control.

Mastra Langfuse HITL Observability MCP